IF ALL LOCAL RINGS ARE STRICTLY HENSELIAN THEN ETALE AND ZARISKI COHOMOLOGY AGREE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the description of the higher direct images, the identification of the stalk through the colimit lemma, and the vanishing over strictly henselian local rings leading to the Leray degeneration.

Editorial note. The label lemma-local-rings-strictly-henselian names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Higher direct images of the comparison morphism as sheafified cohomology. Same chapter, the description of R^p epsilon_*F as the sheaf associated to U |-> H^p_etale(U, F). Supplies the computation of the stalks of the higher direct images.
- Cohomology of a filtered colimit of schemes. Same chapter, the lemma computing the stalk of the sheafified cohomology as the cohomology of Spec of the local ring with pulled-back coefficients. Supplies the identification of the stalk as H^p_etale(Spec(O_{X,x}), G_x).
- Vanishing of etale cohomology over a strictly henselian local ring. Same chapter, the lemma stating the vanishing of the higher etale cohomology of a strictly henselian local ring. Supplies the vanishing that makes the higher direct images zero.

Source reference. Original reference 03Q6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7464–7464. Prerequisite pointer invoked at proof line(s) 14220; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 03QO: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8207–8207. Prerequisite pointer invoked at proof line(s) 14223; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme all of whose local rings are strictly henselian. Then for any abelian sheaf $\mathcal{F}$ on $S_{étale}$ we have $H^i(S_{étale}, \mathcal{F}) = H^i(S_{Zar}, \mathcal{F})$.*

Proof. Let $\epsilon : S_{étale} \to S_{Zar}$ be the morphism of sites given by the inclusion functor. The Zariski sheaf $R^p\epsilon_*\mathcal{F}$ is the sheaf associated to the presheaf $U \mapsto H^p_{étale}(U, \mathcal{F})$. Thus the stalk at $x \in X$ is $\operatorname{colim} H^p_{étale}(U, \mathcal{F}) = H^p_{étale}(\operatorname{Spec}(\mathcal{O}_{X,x}), \mathcal{G}_x)$ where $\mathcal{G}_x$ denotes the pullback of $\mathcal{F}$ to $\operatorname{Spec}(\mathcal{O}_{X,x})$, see Lemma 03Q6. Thus the higher direct images of $R^p\epsilon_*\mathcal{F}$ are zero by Lemma 03QO and we conclude by the Leray spectral sequence. □